



St Catherine's School

Year: 12

Subject: Extension II Mathematics

Time Allowed: 3 hours

Date: August 2005

Exam number: 15227508

Directions to candidates:

- All questions are to be attempted.
- All questions are of equal value.
- All necessary **working** must be shown in every question.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators and geometrical instruments are required.
- Start each question on a new page.
- Hand in your work in 1 bundle:
- Attach the question paper

TEACHER'S USE ONLY	
Total Marks	
A	15 + 15 + 15 + 12 = 57
B	14 + 8 + 14 + 15 = 51
TOTAL	108

Extension II Trials

Question 1

(a) Integrate $\int \frac{dx}{x \ln x}$ (2m)

(b) (i) Show that $z\bar{z} = |z|^2$ (1m)

(ii) hence or otherwise find and sketch the locus of z :

$$(z-2)(\bar{z}-2) = 4 \quad (3m)$$

(c) Integrate $\int \frac{2x}{x^2 + 4x + 5} dx$ (4m)

(d) Find the modulus and argument of $z = 1 + i$ and $w = -1 - \sqrt{3}i$ and

hence the modulus and argument of $\frac{z^2}{w^5}$

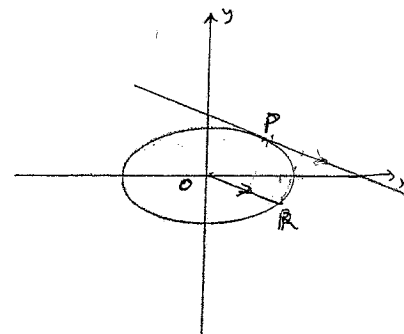
(5m)

Question 2

(a) Show that $x^3 + ax + b = 0$ has a double root if $4a^3 + 27b^2 = 0$ (4m)

(b) $(1+i)$ is a root of the polynomial equation $z^4 - 4z^3 + 10z^2 - 12z + 8 = 0$
Find all the other roots. (4m)

(c) $P(4 \cos \theta, 3 \sin \theta)$ is a point on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.



(i) Show that the gradient of the tangent at P is $\frac{-3 \cos \theta}{4 \sin \theta}$ (2m)

OR is drawn parallel to the tangent from the origin O meeting the ellipse again at the point R as shown.

(ii) Find the equation of OR. (1m)

(iii) Show that the coordinates of R is given by $(4 \sin \theta, -3 \cos \theta)$ (2m)

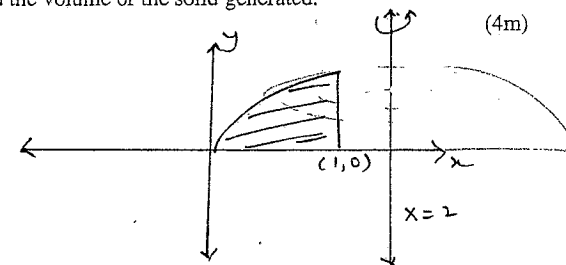
(iv) Find the area of the triangle OPR. (2m)

Question 3

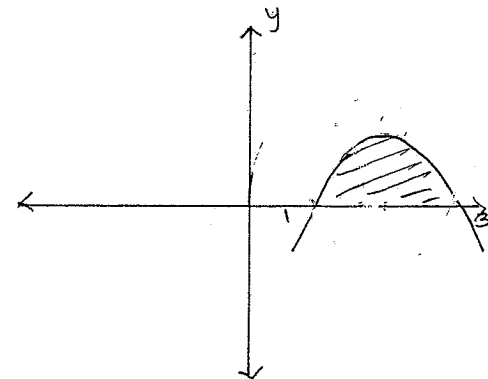
- (a) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on the rectangular hyperbola $xy = c^2$
- (i) Show that the equation of the chord PQ is $x+pqy=cp+cq$ (2m)
- (ii) Find the locus of the mid point of the chord, given that the chord passes through the point (2, 0) (3m)
- (b) Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$
- (i) Find the eccentricity (1m)
- (ii) Find the length of the major axis (1m)
- (iii) Find the coordinates of the foci (1m)
- (iv) Find the equation of the directrices (1m)
- (v) Sketch the ellipse, showing the above features. (1m)
- (c) A particle of mass 5 kilograms attached by a string of length 2 m. to a fixed point describes a horizontal circle with an angular velocity of 3 radians per second.
- (i) Copy the diagram and show the forces acting on the particle (1m)
- (i) Find the tension in the string (3m)
- (ii) Find the distance of the particle below the fixed point. (2m)

Question 4

- (a)(i) A polynomial $P(x)$ is divided by $x^2 - a^2$ and the remainder is $px+q$. Show that
- $$p = \frac{1}{2a}(P(a) - P(-a)) \text{ and } q = \frac{1}{2}(P(a) + P(-a)) \quad (3m)$$
- (ii) Hence or otherwise find the remainder when the polynomial $x^n - a^n$ is divided by $x^2 - a^2$, for when n is even and when n is odd. (3m)
- (b) The area bounded by the curve $y = \sqrt{x}$, the x axis and the line $x=1$ is rotated about the the line $x=2$. By considering the slices perpendicular to the axis of rotation, find the volume of the solid generated. (4m)



- (c) The area bounded by the parabola $y = (x-1)(3-x)$ and the x axis is rotated about the x axis. Use the method of cylindrical shells to find the volume of the solid generated. (5m)

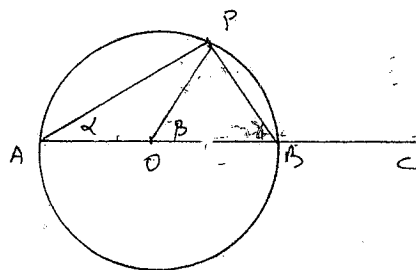


Question 5.

(a) In a bag there are three counters marked with the digit '3' and five counters marked with the digit '4'. Four counters are drawn out of the bag one at a time with replacement.

- (i) Find the probability that three '3's and one '4' will be drawn (2m)
- (ii) Find the probability that the sum of the digits on the counters is greater than 15. (2m)

(c) A particle P is moving in a circle of radius a , with uniform speed u . AB is a diameter of the circle and O is the centre. AB is produced to C (fig). Let angle PAC = α , angle POC = β and PBC = γ . Find, in terms of u , the angular velocity of the particle P about each of the points A, O and B. (4m)



(c) Express $\frac{x^2 + x - 28}{x^2 - 16}$ as a sum of partial fractions. (3m)

(d) (i) Use De Moivre's theorem to show that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ (1m)

(ii) Hence solve $8x^3 - 6x - 1 = 0$ (3m)

Question 6

(a) Let w be a complex root $z^3 = 1$

(i) Show that $1 + w + w^2 = 0$ (2m)

(ii) Find the possible values of w^k , where k is any positive integer. (2m)

(iii) Hence explain why the possible values of $w^k + w^{2k}$ is 2 or -1 (1m)

(iv) Write down the expansions of $(1 + w)^n$ and $(1 + w^2)^n$ (1m)

(v) Deduce that $(1 + w)^n + (1 + w^2)^n = 3({}^nC_0 + {}^nC_3 + {}^nC_6 + \dots + {}^nC_{3l}) - 2^n$, where l is the largest integer such that $3l \leq n$

(use the identity $\sum_{r=0}^n {}^nC_r = 2^n$) (3m)

(vi) If n is a multiple of 6, deduce that

$${}^nC_0 + {}^nC_3 + {}^nC_6 + \dots + {}^nC_n = \frac{1}{3}(2^n + 2) \quad (2m)$$

(b) If $I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx$, $n = 1, 2, 3, \dots$, show that

$$2n I_{n+1} = (2n-1)I_n + \frac{1}{2^n}, \text{ for } n=1, 2, 3, \dots$$

(4m)

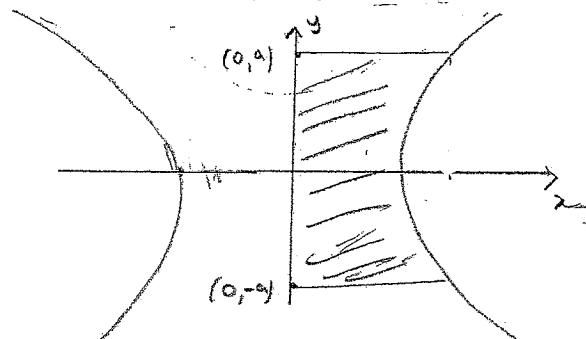
Question 7

- (a) (i) Using integration by parts or otherwise show that

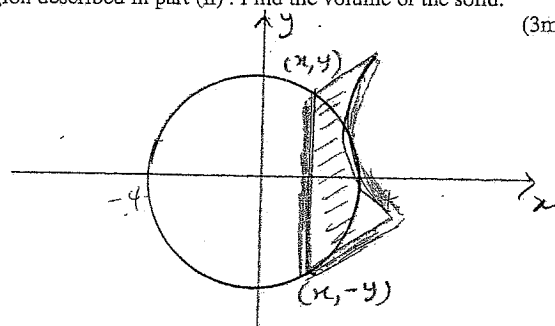
$$\int_0^{\frac{\pi}{4}} \sec^3 \theta \, d\theta = \frac{1}{2}(\sqrt{2} + \ln(\sqrt{2} + 1))$$

(3m)

- (ii) In the given diagram the shaded region R is bounded by a branch of the Hyperbola $x^2 - y^2 = a^2$, the lines $y=a$ and $y=-a$ and the y axis. Show that the area of the region is given by $a^2(\sqrt{2} + \ln(\sqrt{2} + 1))$ (4m)



- (iii) In the diagram a solid is constructed with base the circle $x^2 + y^2 = 16$. Each cross section perpendicular to the x axis is a plane similar to the region described in part (ii). Find the volume of the solid. (3m)



- (b) Assume that the tides rise and fall in Simple Harmonic Motion. At low tide a channel is 9 metres deep and at high tide, it is 12 metres deep. The low tide occurs at 9 am and the high tide at 3pm. A ship needs 11.25 metres of water depth to pass through. Find the earliest time between 9 am and 3 pm, when it is safe for the ship to pass through

(5m)

Question 8

- (a) A body of mass 1 kilogram is projected upwards from the ground at 20 metres per second. The particle is under the effect of gravity and the air resistance at any time is equal to $\frac{1}{10}v^2$, where v is the velocity at the time.

(Take the acceleration due to gravity at 10.)

- (i) Explain why the equation of motion while going up is given by

$$\ddot{x} = -(10 + \frac{1}{10}v^2)$$

(1m)

- (ii) Taking $\ddot{x} = v \frac{dv}{dx}$ find the greatest height reached

(3m)

- (iii) Taking $\ddot{x} = \frac{dv}{dt}$, find the time taken to reach this height.

(3m)

- (b) (i) If $I_{2n} = \int_0^{\frac{\pi}{2}} \cos^{2n} x \, dx$, use integration by parts to show that

$$I_{2n} = \frac{2n-1}{2n} I_{2n-2} \quad (4m)$$

- (ii) Show that $I_4 = \frac{3\pi}{16}$ (1m)

- (iii) Show that $I_{2n} = \frac{(2n)!}{(2^n n!)^2} \frac{\pi}{2}$ (3m)

End of Paper

Q.1

$$1. \int \frac{dx}{x \ln x} \quad \text{if } \ln x = u$$

$$\frac{1}{x} dx = du$$

$$= \int \frac{du}{u}$$

$$= \ln u + C$$

$$= \ln(\ln x) + C$$

2. $z\bar{z} = |z|^2$

Let $z = x + iy$

$\bar{z} = x - iy$

$$z\bar{z} = (x+iy)(x-iy)$$

$$= x^2 + y^2$$

$$|z|^2 = (\sqrt{x^2+y^2})^2$$

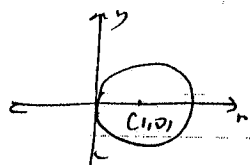
$$= x^2 + y^2$$

$$(z-2)(\bar{z}-2) = 4$$

$$z\bar{z} - 2(z+\bar{z}) + 4 = 4$$

$$x^2 + y^2 - 2x = 0$$

$$(x-1)^2 + y^2 = 1$$



Circle centre (1,0)
 $r = 1$

c) $\int \frac{2x}{x^2+4x+5} dx$

$$= \int \frac{2x+4-4}{x^2+4x+5} dx$$

$$= \ln(x^2+4x+5) - \int \frac{4}{(x+2)+1} dx$$

$$= \ln(x^2+4x+5) - 4 \ln|x+2| + C$$

d) $z = 1+i$

$$|z| = \sqrt{2} \quad ; \quad \arg z = \frac{\pi}{4}$$

$$w = -1 - \sqrt{3}i$$

$$|w| = 2 \quad \arg w = -\frac{2\pi}{3}$$

$$\left| \frac{z^2}{w^3} \right| = \frac{(\sqrt{2})^2}{2^3} = \frac{1}{2}$$

$$\arg \frac{z^2}{w^3} = 2\arg z - 3\arg w \pm 2n\pi$$

$$= \frac{\pi}{2} + \frac{10\pi}{3} (\pm 2n\pi)$$

$$= -\frac{\pi}{6}$$

Q.2

$$x^3 + ax + b = 0$$

if a is the double root

a is also a root of

$$3x^2 + a = 0$$

Thus

$$a^3 + aa + b = 0 \quad \text{--- (1)}$$

$$3a^2 + a = 0 \quad \text{--- (2)}$$

from (1)

$$a(a^2 + a) = -b$$

$$a^2(a^2 + a) = -b^2$$

$$-\frac{a}{3} \left(-\frac{a}{3} + a \right)^2 = -b^2 \quad \text{from (2)}$$

$$-\frac{a}{3} \times \frac{4a^2}{9} = -b^2$$

$$-4a^3 = 27b^2$$

$$\text{or } 4a^3 + 27b^2 = 0$$

(4) $z^4 - 4z^3 + 10z^2 - 12z + 8 = 0$

— roots have real coefficients.

$\therefore$ if $(1+i)$ is a root, $(1-i)$ is also a root.

Let z and w be the other roots

(Note: if they are complex roots, they are also complex conjugates.)

$$(z-(1+i))(z-(1-i)) \text{ is a factor.}$$

$$z^2 - 2z + 2 \text{ is a factor.}$$

$$\begin{array}{r} z^2 - 2z + 2 \overline{) z^4 - 4z^3 + 10z^2 - 12z + 8} \\ \underline{z^4 - 2z^3 + 2z^2} \\ -2z^3 + 8z^2 - 12z + 8 \\ \underline{-2z^3 + 4z^2 - 4z} \\ 4z^2 - 8z + 8 \end{array}$$

$$4z^2 - 8z + 8$$

Thus $(z^2 - 2z + 2)(z^2 - 2z + 4) = 0$

$$\therefore z = \frac{1+i}{1-i} \quad z = \frac{2 \pm 2\sqrt{3}i}{2}$$

$$\text{ie } z = 1 + \sqrt{3}i$$

$$z = 1 - \sqrt{3}i$$

c)

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$\frac{2x}{16} + \frac{2y}{9} y' = 0$$

$$y' = -\frac{x}{16} \div \frac{y}{9}$$

$$= -\frac{9x}{16y}$$

$$y'_{\text{at } P} = -\frac{9}{16} \cdot \frac{4 \cos \theta}{3 \sin \theta}$$

$$= -\frac{3 \cos \theta}{4 \sin \theta}$$

Eqn. of OR: $y = -\frac{3 \cos \theta}{4 \sin \theta} x$

(11) $\frac{x^2}{16} + \frac{9 \cos^2 \theta}{16 \sin^2 \theta} \cdot \frac{x^2}{9} = 1$

$$x^2 \sin^2 \theta + x^2 \cos^2 \theta = 16 \sin^2 \theta$$

$$x = \pm 4 \sin \theta$$

where R is $x = 4 \sin \theta$

$$\therefore y = -\frac{3 \cos \theta}{4 \sin \theta} \cdot 4 \sin \theta = -3 \cos \theta$$

Thus R: $(4 \sin \theta, -3 \cos \theta)$

(iv) $|OR| = \sqrt{16 \sin^2 \theta + 9 \cos^2 \theta}$

h dist. from P: $(4 \cos \theta, 3 \sin \theta)$ is

$$y = -\frac{3 \cos \theta}{4 \sin \theta} x$$

or $3 \cos \theta x + 4 \sin \theta y = 0$ is.

$$d = \frac{|12 \cos^2 \theta + 12 \sin^2 \theta|}{\sqrt{16 \sin^2 \theta + 9 \cos^2 \theta}} = \frac{12}{\sqrt{16 \sin^2 \theta + 9 \cos^2 \theta}}$$

$\therefore$ Area is $\frac{1}{2} \times |OR| \times d$

$$= \frac{1}{2} \times \sqrt{\frac{12}{\sqrt{16 \sin^2 \theta + 9 \cos^2 \theta}}}$$

$$= 6 \text{ unit}^2$$

Q.3. $P: (cp, \frac{c}{p})$; $Q: (cq, \frac{c}{q})$

grad. of PA: $\frac{\frac{c}{q} - \frac{c}{p}}{cq - cp} = \frac{p-q}{pq} \times \frac{1}{q-p} = -\frac{1}{pq}$

Eqn. of PA: $y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$

$pqy - cq = -x + cp$
 $x + pqy = c(p+q)$

passes through $(2,0)$ $\therefore 2 = c(p+q)$ — (1)

mid. pt. of PA:

$x = c\frac{(p+q)}{2}$ $y = \frac{1}{2}(\frac{c}{p} + \frac{c}{q})$

$= \frac{2}{2}$

from (1)

$= 1$ is the locus.

(b). $\frac{x^2}{9} + \frac{y^2}{16} = 1$

$9 = 16(1 - e^2)$

$1 - e^2 = \frac{9}{16}$

$e^2 = \frac{7}{16}$ $e = \frac{\sqrt{7}}{4}$ (esp)

foci: $(0, \pm ae)$

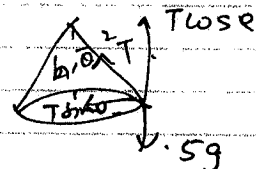
$= (0, \pm \sqrt{7})$ —

major axis = 8.

directrices: $y = \pm 4 \times \frac{4}{\sqrt{7}}$

$y = \pm \frac{16}{\sqrt{7}}$

c)



$\omega = 3 \text{ rad/s}$

$T \cos \theta = 5g$ — (1)

$T \sin \theta = 5 \times 2 \sin \theta \times 9$

$T = 90 \text{ N}$

Sub in (1)

$90 \cos \theta = 5g$

$\cos \theta = \frac{5g}{90}$ also: $\cos \theta = \frac{h}{2}$

$\therefore \frac{h}{2} = \frac{5g}{90}$

$h = \frac{10g}{90} = \frac{9}{9} \text{ m}$

Q.4

$P(x) = (x^2 - a^2) \cdot Q(x) + px + q$

$P(a) = ap + q$ — (1)

$P(-a) = -ap + q$ — (2)

$\frac{P(a) + P(-a)}{2} = q$ $\frac{P(a) - P(-a)}{2a} = p$

The rem:

$P(x): x^n - a^n$

$P(a) = a^n - a^n = 0$

$$\begin{aligned}
 P(-a) &= (-a)^n - a^n \\
 &= (-1)^n a^n - a^n \\
 &= a^n - a^n \quad \text{when } n \text{ is even.} \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 &= (-1)^n a^n - a^n \\
 &= -a^n - a^n \quad \text{when } n \text{ is odd.} \\
 &= -2a^n
 \end{aligned}$$

$\therefore$ The rem. is $px + q$, where
when n is even.

$$p = \frac{1}{2a} (P(a) - P(-a))$$

$$= 0.$$

$$q = 0$$

The rem. is zero

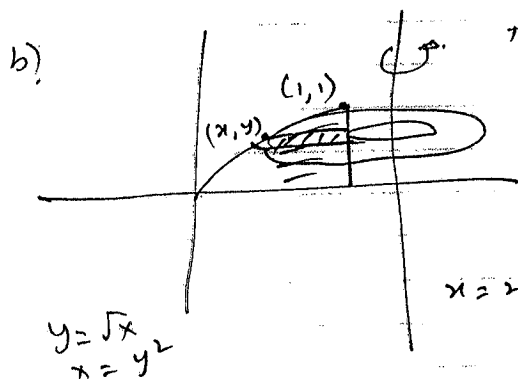
when n is odd.

$$p = \frac{1}{2a} (0 - (-2a^n))$$

$$= a^{n-1}.$$

$$q = \frac{1}{2} (-2a^n) = -a^n.$$

$\therefore$ The rem. is $a^{n-1}x - a^n$.



$$\Delta V = \pi (2-x)^2 - 1^2 \Delta y$$

$$= \pi (4 - 4x + x^2 - 1)$$

$$= \pi (3 - 4x + x^2) \Delta y$$

$$= \pi (3 - 4y^2 + y^4) \Delta y.$$

$$V = \pi \int_0^1 (3 - 4y^2 + y^4) dy$$

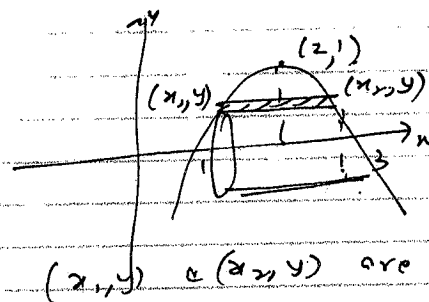
$$= \pi \left(3y - \frac{4}{3} y^3 + \frac{y^5}{5} \right) \Big|_0^1$$

$$= \pi \left(3 - \frac{4}{3} + \frac{1}{5} \right) u^3.$$

$$= \pi \left(\frac{45 - 20 + 3}{15} \right) u^3.$$

$$= \frac{28\pi}{15} u^3.$$

c)



consider a cyl. shell.

at a dist. y from the
axis of rotation.

$$\Delta V = 2\pi y (x_2 - x_1) \Delta y$$

$$y = (x-1)(3-x)$$

$$= -x^2 + 4x - 3$$

x_1, x_2 are roots of

$$x^2 - 4x + 3 + y = 0$$

$$\therefore x_1 + x_2 = 4$$

$$x_1, x_2 = 3 + y.$$

$$(x_1 - x_2)^2 = 4^2 - 4(3+y)$$

$$= 16 - 12 - 4y$$

$$= 4 - 4y$$

$$\therefore x_1 - x_2 = \sqrt{4 - 4y}$$

$$x_1, x_2 = 2\sqrt{1-y}$$

$$\therefore \Delta V = 2\pi y (2\sqrt{1-y})$$

$$= 4\pi y \sqrt{1-y} \Delta y$$

$$\therefore V = 4\pi \int_0^1 y \sqrt{1-y} dy$$

$$\text{let } 1-y = u.$$

$$-dy = du.$$

$$y=0; u=1$$

$$y=1; u=0$$

$$\begin{aligned}
 \therefore V &= 4\pi \int_0^1 (1-u) \sqrt{u} (-du) \\
 &= 4\pi \int_0^1 (\sqrt{u} - u^{3/2}) du \\
 &= 4\pi \left[\frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^1 \\
 &= 4\pi \left(\frac{2}{3} - \frac{2}{5} \right) \\
 &= 8\pi \times \frac{2}{15} = \frac{16\pi}{15} u^3.
 \end{aligned}$$

Q.5

$$P(3) = \frac{3}{8} = p$$

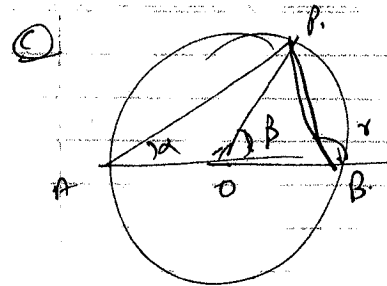
$$P(4) = \frac{5}{8} = q$$

4. Counters

$$\begin{aligned}
 \text{Write } (p+q)^4 &= 4C_0 p^4 + 4C_1 p^3 q + 4C_2 p^2 q^2 + 4C_3 p q^3 + 4C_4 q^4 \\
 P(3 \text{ 3's and one 4}) &= 4C_1 \left(\frac{3}{8}\right)^3 \left(\frac{5}{8}\right) \\
 &= 4 \times \frac{27 \times 5}{8^4}
 \end{aligned}$$

① For sum > 15, we need to draw 4 4's.

$$\begin{aligned}
 P(4 \text{ 4's}) &= 4C_0 \left(\frac{5}{8}\right)^4 \\
 &= \left(\frac{5}{8}\right)^4.
 \end{aligned}$$



Speed: u .
radius: a .

$$u = a\omega$$

$$\therefore \omega = \frac{u}{a}$$

$$\therefore \beta = \frac{u}{a}$$

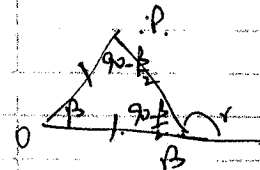
β is ext. angle of the isos $\triangle OAP$

$$\therefore \beta = 2\alpha$$

$$\therefore \alpha = \frac{\beta}{2}$$

$$\therefore \alpha = \frac{1}{2} \beta = \frac{1}{2} \frac{u}{a}$$

γ is an ext. $\angle$ of the isos $\triangle OPB$.



$$\gamma = 90 + \frac{\beta}{2}$$

$$\therefore \gamma = \frac{1}{2} \beta = \frac{1}{2} \frac{u}{a}$$

$$\begin{array}{r}
 1 \\
 \hline
 (x^2 - 16) \overline{) x^2 + x - 28} \\
 \underline{x^2 } -16 \\
 \hline
 x - 12
 \end{array}$$

$$\therefore \frac{x^2 + x - 28}{x^2 - 16} = 1 + \frac{x - 12}{x^2 - 16}$$

$$\text{Consider } \frac{x-12}{(x-4)(x+4)} = \frac{A}{x-4} + \frac{B}{x+4}$$

$$\therefore x-12 = A(x+4) + B(x-4)$$

$$\text{Let } x=4$$

$$-8 = 8A \quad A = -1$$

$$\text{Let } x = -4; \quad -16 = -8B \\ B = 2$$

$$\therefore \frac{x^2 + x - 28}{x^2 - 16} = 1 - \frac{1}{x-4} + \frac{2}{x+4}$$

1) Consider,

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta \quad \text{by De Moivre's th.}$$

also

$$(\cos \theta + i \sin \theta)^3 = 3\cos^3 \theta + 3\cos^2 \theta (i \sin \theta) + 3\cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$$

equating real parts

$$\begin{aligned} \therefore \cos 3\theta &= \cos^3 \theta - 3\cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3\cos \theta (1 - \cos^2 \theta) \\ &= 4\cos^3 \theta - 3\cos \theta \end{aligned}$$

$$\text{if } \cos 3\theta = \frac{1}{2}; \quad 4\cos^3 \theta - 3\cos \theta = \frac{1}{2} \\ \text{or } 8\cos^3 \theta - 6\cos \theta - 1 = 0$$

hence if α, β, γ are the roots of the

above eqn.

$\cos \alpha, \cos \beta, \cos \gamma$ would be the solution to

$$8x^3 - 6x - 1 = 0$$

α, β, γ are the roots to $\cos 3\theta = \frac{1}{2}$

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$\therefore$ The roots of $8x^3 - 6x - 1 = 0$ are

$$\cos \frac{\pi}{9}, \cos \frac{5\pi}{9} \text{ and } \cos \frac{7\pi}{9}$$

Q.6

$$z^3 = 1$$

$$z^3 - 1 = 0$$

$$(z-1)(z^2 + z + 1) = 0$$

$$z = 1; \quad z = \frac{-1 \pm \sqrt{3}i}{2}$$

$$\text{if } w = \frac{-1 + \sqrt{3}i}{2}, \quad w^2 = \frac{(-1 + \sqrt{3}i)^2}{2^2} = \frac{-1 - \sqrt{3}i}{2}$$

Thus $1, w, w^2$ are the roots of $z^3 - 1 = 0$

$$\therefore 1 + w + w^2 = 0$$

$$\text{if } k = 3m; \quad w^k = w^{3m} = (w^3)^m = 1^m = 1$$

$$\text{if } k = 3m+1; \quad w^k = w^{3m+1} = w^{3m} \cdot w = 1 \cdot w = w$$

$$\text{if } k = 3m+2; \quad w^k = w^{3m+2} = w^{3m} \cdot w^2 = 1 \cdot w^2 = w^2$$

$\therefore$ The possible values are $1, w, w^2$

11) $w^k + w^{2k} = 1 + 1 = 2$ if $k = 3m$

$$w^k + w^{2k} = w + w^2 = -1 \quad \text{if } k = 3m+1$$

$$w^k + w^{2k} = w^2 + w^4 = w^2 + w = -1 \quad \text{if } k = 3m+2$$

$$v) \quad (1+w)^n = n_{c0} + n_{c1} w + n_{c2} w^2 + \dots + n_{cn} w^n.$$

$$(1+w^2)^n = n_{c0} + n_{c1} w^2 + n_{c2} w^4 + \dots + n_{cn} w^{2n}.$$

$$vi) \quad (1+w)^n + (1+w^2)^n$$

$$= 2n_{c0} + n_{c1}(w+w^2) + n_{c2}(w^2+w^4) + n_{c3}(w^3+w^6)$$

$$+ \dots + n_{cn}(w^n + w^{2n}),$$

$$= 2n_{c0} - n_{c1} - n_{c2} + 2n_{c3} - n_{c4} - n_{c5} + 2n_{c6} \dots$$

$$= 2(n_{c0} + n_{c3} + n_{c6} + \dots + n_{c_{3k}}) - (n_{c1} + n_{c2} + n_{c4} + \dots)$$

$$= 3(n_{c0} + n_{c3} + n_{c6} + \dots + n_{c_{3k}}) - \sum_{r=0}^n n_{cr}$$

$$= 3(n_{c0} + n_{c3} + \dots + n_{c_{3k}}) - 2^n.$$

$$vii) \quad \text{if } n \text{ is a multiple of } 6;$$

$$(1+w)^n = (-w^2)^{6k} = w^{12k} = (w^3)^{4k} = 1^{4k} = 1$$

$$(1+w^2)^n = (-w)^{6k} = w^{6k} = (w^3)^{2k} = 1^{2k} = 1.$$

$$\therefore (1+w)^n + (1+w^2)^n = 2.$$

$$\text{Thus } 2 = 3(n_{c0} + n_{c3} + \dots + n_{c_n}) - 2^n$$

$$3(n_{c0} + n_{c3} + \dots + n_{c_n}) = 2 + 2^n.$$

$$\text{or } n_{c0} + n_{c3} + \dots + n_{c_n} = \frac{1}{3}(2^n + 2).$$

$$b) \quad I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx.$$

$$\text{Let } u = (1+x^2)^{-n} \quad v' = 1$$

$$u' = -n(1+x^2)^{-n-1} (2x) \quad v = x$$

$$= -\frac{2nx}{(1+x^2)^{n+1}}.$$

$$\int u v' = uv - \int u' v.$$

$$\therefore I_n = \left(\frac{x}{(1+x^2)^n} \right)_0^1 + 2n \int_0^1 \frac{x^2}{(1+x^2)^{n+1}} dx$$

$$= \frac{1}{2^n} + 2n \left[\int_0^1 \frac{x^2+1-1}{(x^2+1)^{n+1}} dx \right]$$

$$I_n = \frac{1}{2^n} + 2n \left[\int_0^1 \frac{1}{(x^2+1)^n} dx - \int_0^1 \frac{1}{(x^2+1)^{n+1}} dx \right]$$

$$\therefore I_n = +2n \left[I_n - I_{n+1} \right] + \frac{1}{2^n}.$$

$$(1-2n) I_n = -2n I_{n+1} + \frac{1}{2^n} \quad \text{or} \quad 2n I_{n+1} = 2n$$

$$\therefore I_n = \frac{2n}{2n+1} I_{n+1} + \frac{1}{2^{n+1}}.$$

Q7. $\int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta$

Let $u = \sec \theta$

$v' = \sec^2 \theta$

$v = \tan \theta$

$u' = \sec \theta \tan \theta$

$\int u v' = uv - \int u' v$

$$\int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta = \left(\sec \theta \tan \theta \right)_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \sec \theta \tan^2 \theta d\theta$$

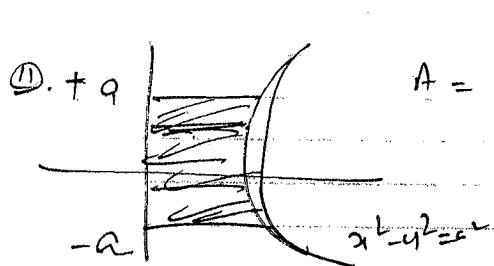
$$= \sqrt{2} - \left(\int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta - \int_0^{\frac{\pi}{4}} \sec \theta d\theta \right)$$

$$I = \sqrt{2} - I + \left(\ln |\sec \theta + \tan \theta| \right)_0^{\frac{\pi}{4}}$$

$$2I = \sqrt{2} + \left(\ln(\sqrt{2}+1) - \ln(1+0) \right)$$

$$= \sqrt{2} + \ln(\sqrt{2}+1)$$

$$I = \frac{1}{2} \left(\sqrt{2} + \ln(\sqrt{2}+1) \right)$$



$$A = \int_{-a}^a x dy$$

$$x^2 - y^2 = a^2$$

$$x^2 = a^2 + y^2$$

$$x = \sqrt{a^2 + y^2}$$

$$A = \int_{-a}^a \sqrt{a^2 + y^2} dy$$

$$= 2 \int_0^a \sqrt{a^2 + y^2} dy \quad (\sqrt{a^2 + y^2} \text{ is an even f.})$$

Let $y = a \tan \theta$

$dy = a \sec^2 \theta d\theta$

$a^2 + y^2 = a^2 + a^2 \tan^2 \theta$

$= a^2 \sec^2 \theta$

$y = 0; \theta = 0$

$y = a; \theta = \frac{\pi}{4}$

$$\therefore A = 2 \int_0^{\frac{\pi}{4}} a \sec \theta \cdot a \sec^2 \theta d\theta$$

$$= 2a^2 \int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta$$

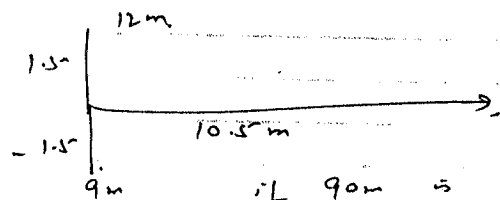
$$= 2a^2 \cdot \left[\frac{1}{2} (\sqrt{2} + \ln(\sqrt{2}+1)) \right]$$

$$= a^2 [\sqrt{2} + \ln(\sqrt{2}+1)]$$

(iii) $\Delta V = y^2 (\sqrt{2} + \ln(\sqrt{2}+1)) \Delta x$

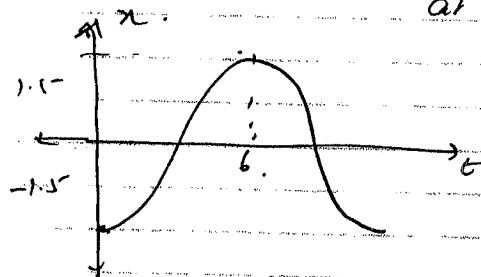
$$V = (\sqrt{2} + \ln(\sqrt{2}+1)) \int_{-4}^4 y^2 dx$$

$$\begin{aligned}
 V &= \sqrt{2 + \ln(\sqrt{2} + 1)} \int_{-4}^4 (16 - x^2) dx \\
 &= 2(\sqrt{2} + \ln(\sqrt{2} + 1)) \int_0^4 (16 - x^2) dx \\
 &= 2(\sqrt{2} + \ln(\sqrt{2} + 1)) \left(16x - \frac{x^3}{3} \right)_0^4 \\
 &= 2(\sqrt{2} + \ln(\sqrt{2} + 1)) \left(64 - \frac{64}{3} \right) \\
 &= \frac{256}{3} (\sqrt{2} + \ln(\sqrt{2} + 1)) \cdot 4^3
 \end{aligned}$$



if 9am is 0 hrs.
3pm : 6 hrs:

Origin: (9am, 10.5m) ; Place the axes at this origin.



The motion can be described by the equation

$$x = -1.5 \cos \frac{2\pi}{n} t$$

where $\frac{2\pi}{n}$ is the period

$$\therefore \frac{2\pi}{n} = 12$$

$$2\pi = 12n$$

$$n = \frac{2\pi}{12} = \frac{\pi}{6}$$

$$\therefore x = -1.5 \cos \frac{\pi}{6} t$$

At 11.25 m depth ; $x = 11.25 - 10.5$
 $= 0.75$

$$\therefore 0.75 = -1.5 \cos \frac{\pi}{6} t$$

$$-\frac{1}{2} = \cos \frac{\pi}{6} t$$

$$\therefore \frac{\pi}{6} t = \frac{2}{3} \pi$$

$$t = 4$$

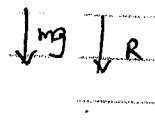
$\therefore$ 4 hrs from 9am i.e. at 1pm is the earliest time when the ship can pass through.

Q.8

Note:

Resistance opposes motion.

if going up is considered positive ;



$$F = ma$$

$$-mg - R = ma$$

$$mg = \frac{1}{10} v^2 = ma$$

m = 1 kg

$$\therefore a = -g - \frac{1}{10} v^2$$

$$= -(10 + \frac{1}{10} v^2)$$

$$v \frac{dv}{dx} = - \frac{100 + v^2}{10}$$

$$\int \frac{10v dv}{100 + v^2} = - \int dx$$

$$\therefore \frac{1}{2} \ln(100 + v^2) = -\frac{1}{10} x + C$$

$$\text{at } x=0; v=20$$

$$\therefore \frac{1}{2} \ln 500 = C$$

$$\therefore \frac{1}{10} x = \frac{1}{2} \ln 500 - \frac{1}{2} \ln(100 + v^2)$$

$$x = 5 \ln 500 - 5 \ln(100 + v^2)$$

$$= 5 \ln \frac{500}{100 + v^2}$$

$$\text{At the greater H; } v=0$$

$$\therefore H = 5 \ln 5$$

$$a) \frac{dv}{dt} = - \frac{100 + v^2}{10}$$

$$\int \frac{dv}{100 + v^2} = - \frac{1}{10} \int dt$$

$$\frac{1}{10} \tan^{-1} \frac{v}{10} = -\frac{1}{10} t + C$$

$$\text{at } t=0; v=20$$

$$\therefore \frac{1}{10} \tan^{-1} 2 = \frac{1}{10} C$$

$$\therefore \frac{1}{10} t = \frac{1}{10} \tan^{-1} 2 - \frac{1}{10} \tan^{-1} \frac{v}{10}$$

$$t = \tan^{-1} 2 - \tan^{-1} \frac{v}{10}$$

$$\text{at the greater H; } v=0$$

$$\therefore t = \tan^{-1} 2 - \tan^{-1} 0$$

$$= \tan^{-1} 2$$

$$b) I_{2n} = \int_0^{\frac{\pi}{2}} \cos^{2n} x \, dx$$

$$\text{Let } u = \cos^{2n-1} x \quad v' = \cos x$$

$$u' = (2n-1) \cos^{2n-2} x (-\sin x) \quad v = \sin x$$

$$\int u v' = u v - \int u' v$$

$$I_{2n} = \int_0^{\frac{\pi}{2}} \cos^{2n-1} x \sin x \, dx + \int_0^{\frac{\pi}{2}} (2n-1) \cos^{2n-2} x \sin^2 x \, dx$$

$$= (2n-1) \int_0^{\frac{\pi}{2}} \cos^{2n-2} x (1 - \cos^2 x) \, dx$$

$$I_{2n} = (2n-1) \left(I_{2n-2} - I_{2n} \right)$$

$$(1+2n-1) I_{2n} = (2n-1) I_{2n-2}$$

$$2n I_{2n} = (2n-1) I_{2n-2}$$

$$I_{2n} = \frac{2n-1}{2n} I_{2n-2}$$

$$I_4 = \frac{3}{4} I_2$$

$$= \frac{3}{4} \cdot \frac{1}{2} \cdot I_0$$

$$= \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$$

$$= \frac{3}{16} \pi$$

$$\begin{aligned} I_0 &= \int_0^{\pi/2} (\cos x)^0 dx \\ &= \int_0^{\pi/2} 1 dx \\ &= \frac{\pi}{2} \end{aligned}$$

$$= \frac{2n(2n-1)(2n-2) \dots 3 \cdot 2 \cdot 1}{2^n \cdot n!} \cdot \frac{\pi}{2}$$

$$= \frac{(2n)!}{(2^n \cdot n!)^2} \cdot \frac{\pi}{2}$$

$$(iii) I_{2n} = \frac{2n-1}{2n} I_{2n-2}$$

$$= \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdot I_{2n-4}$$

$$= \frac{(2n-1)}{2n} \cdot \frac{(2n-3)}{2n-2} \cdot \frac{2n-5}{2n-4} \cdot \frac{2n-7}{2n-6} \dots I_4$$

$$= \frac{(2n-1)}{2n} \cdot \frac{(2n-3)}{2(n-1)} \cdot \frac{(2n-5)}{2(n-2)} \cdot \frac{(2n-7)}{2(n-4)} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= \frac{(2n-1) \cdot (2n-3) \dots 3 \cdot 1}{2^n \cdot n!} \cdot \frac{2n(2n-2)(2n-4) \dots 2}{2^n(2n-2)(2n-4) \dots 2} \cdot \frac{\pi}{2}$$